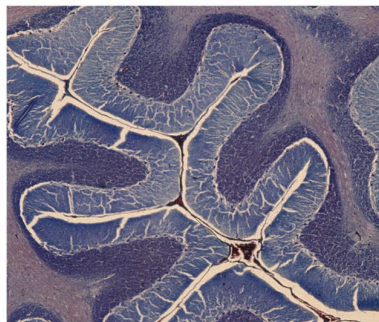


BRAIN

THE BRAIN, TOGETHER WITH THE SPINAL CORD, REGULATES NON-CONSCIOUS PROCESSES AND COORDINATES MOST VOLUNTARY MOVEMENT. IT IS THE SITE OF CONSCIOUSNESS, ALLOWING HUMANS TO THINK AND LEARN.

BRAIN STRUCTURE

The largest part of the brain is the cerebrum, which has a heavily folded surface – the pattern of which is unique in each person. The grooves are called sulci when shallow and fissures when deep. Fissures and some of the large sulci outline four functional areas, called lobes: frontal, parietal, occipital, and temporal (see p.90). A ridge on the surface of the brain is called a gyrus. The centre of the brain contains the thalamus, which acts as the brain's information relay station. Surrounding this is a group of structures known as the limbic system (see p.94), which is involved in survival instincts, behaviour, and emotions. Closely linked with the limbic system is the hypothalamus (see p.95), which receives sensory information.



CEREBELLUM

The cerebellum (section shown above) contains billions of neurons that link up with other regions of the brain and spinal cord to facilitate precise movement.

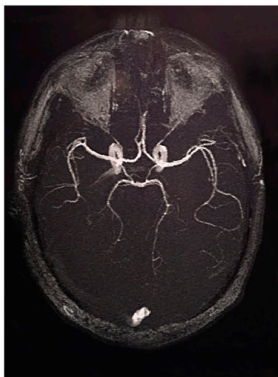
BLOOD SUPPLY TO THE BRAIN

The brain forms 2 per cent of the body's weight but needs 20 per cent of its blood. Without oxygen and glucose, brain function quickly deteriorates, leading to dizziness and loss of consciousness. Within only four to eight minutes of oxygen deprivation, brain

damage, or death, results. The brain has an abundant supply of blood from a vast network of blood vessels that stem from the carotid arteries, which run up each side of the neck, and from two vertebral arteries that run alongside the spinal cord.

CIRCLE OF WILLIS

A ring of arteries, the Circle of Willis, encircles the base of the brain and provides multiple pathways to supply oxygenated blood to all parts of the brain. If a pathway becomes blocked, blood can be redirected from another pathway.



BLOOD SUPPLY

The brain has an extensive blood supply from two front and two rear arteries, as illustrated in this colour, three-dimensional magnetic resonance angiography (MRA) scan. The blood vessels are coloured in red; here, they are seen supplying oxygenated blood to various parts of the brain, which is shown as the blue area.

